

BIOLOGY (THEORY) – MOCK TEST 3

Boards Dominator Practice Paper • CBSE Pattern Trends

Time: 3 Hours

Max Marks: 70

By Pragya Thunderstudy

SECTION A (16 Marks)

Q1. The histone core in nucleosome is made up of:

- a) Pentamer
- b) Hexamer
- c) Octamer
- d) Heptamer

Q2. Colostrum secreted during early lactation is rich in:

- a) IgE
- b) IgD
- c) IgA
- d) IgG

Q3. During transcription, RNA polymerase uses:

- a) dNTPs
- b) Ribonucleotide triphosphates
- c) Deoxyribonucleosides
- d) Ribonucleosides

Q4. DNA is separated in agarose gel electrophoresis using:

- a) Visible light
- b) Infrared light
- c) UV light + Ethidium bromide
- d) Blue light

Q5. Which enzyme joins Okazaki fragments?

- a) Helicase
- b) DNA ligase
- c) Primase
- d) Topoisomerase

Q6. Cytochrome-C is important evidence for:

- a) Genetic drift
- b) Evolutionary relationship
- c) Mutation
- d) Isolation

Q7. Pollination in Vallisneria occurs:

- a) By wind
- b) On the water surface
- c) By insects
- d) By birds

Q8. The following is an example of codominance:

- a) ABO blood group
- b) Height
- c) Skin colour
- d) Seed shape

Q9. Recombinant DNA is formed using:

- a) Helicase
- b) Restriction enzyme + Ligase
- c) Polymerase only
- d) Primase

Q10. Down syndrome is caused due to:

- a) Trisomy 21
- b) Monosomy
- c) Deletion mutation
- d) Duplication

Q11. DNA isolation from fungal cell requires:

- a) Cellulase
- b) Chitinase
- c) Lysozyme
- d) Protease

Q12. The process where male insect mistakes flower as female partner is called:

- a) Pseudocopulation
- b) True pollination
- c) Autogamy
- d) Cleistogamy

Q13. The outermost layer of uterus is called:

- a) Endometrium
- b) Myometrium
- c) Perimetrium
- d) Epithelium

Q14. Interferons are secreted by:

- a) RBC
- b) Virus infected cells
- c) Bacteria
- d) Platelets

Q15. The most common host in recombinant DNA technology is:

- a) Yeast
- b) E. coli
- c) Virus
- d) Plant cells

Q16. Biogas plants are common in rural areas because:

- a) Dung is rich in Methanogens
- b) Less electricity
- c) More pollution
- d) Soil fertility

SECTION B (10 Marks)

Q17. Explain structure of antibody molecule with labelled diagram.

Q18. Study DNA replication fork diagram and answer:

- a) Identify the correct template polarity for lagging strand synthesis.
- b) Name the enzyme joining DNA fragments.

Q19. What is amniocentesis? How is it misused?

Q20. Explain interaction between fig tree and wasp as mutualism.

Q21. Draw ecological pyramid of numbers for: 32 birds ← 20 insects ← 1 tree.

SECTION C (21 Marks)

Q22. Explain neuroendocrine mechanism of parturition.

Q23. A child has extra chromosome 21. a) Name disorder b) Write symptoms.

Q24. Define BOD and explain its importance.

Q25. Write three advantages of genetically modified plants.

Q26. Explain inheritance pattern of ABO blood group with a genotype table.

Q27. Explain habitat fragmentation and extinction with an example.

Q28. Study outbreeding device diagram and answer: a) Advantage of cross pollination b) Can plant show geitonogamy?

SECTION D (8 Marks)

Q29. Case Study: Cytochrome-C is used to trace evolution.

- a) Why conserved proteins help evolution study?
- b) Why humans and chimpanzees show similarity?
- c) How molecular homology supports evolution?

Q30. Case Study: Recombinant DNA technology flow diagram.

- a) Name enzyme multiplying foreign DNA.
- b) How EcoRI helps ligase.
- c) Name host organism.

SECTION E (15 Marks)

Q31. Draw labelled diagram of human sperm and write functions of any four parts.

Q32. Draw stages of embryo development in dicot plant.

Q33. Explain Hershey and Chase experiment with labelled diagram and conclusion.

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